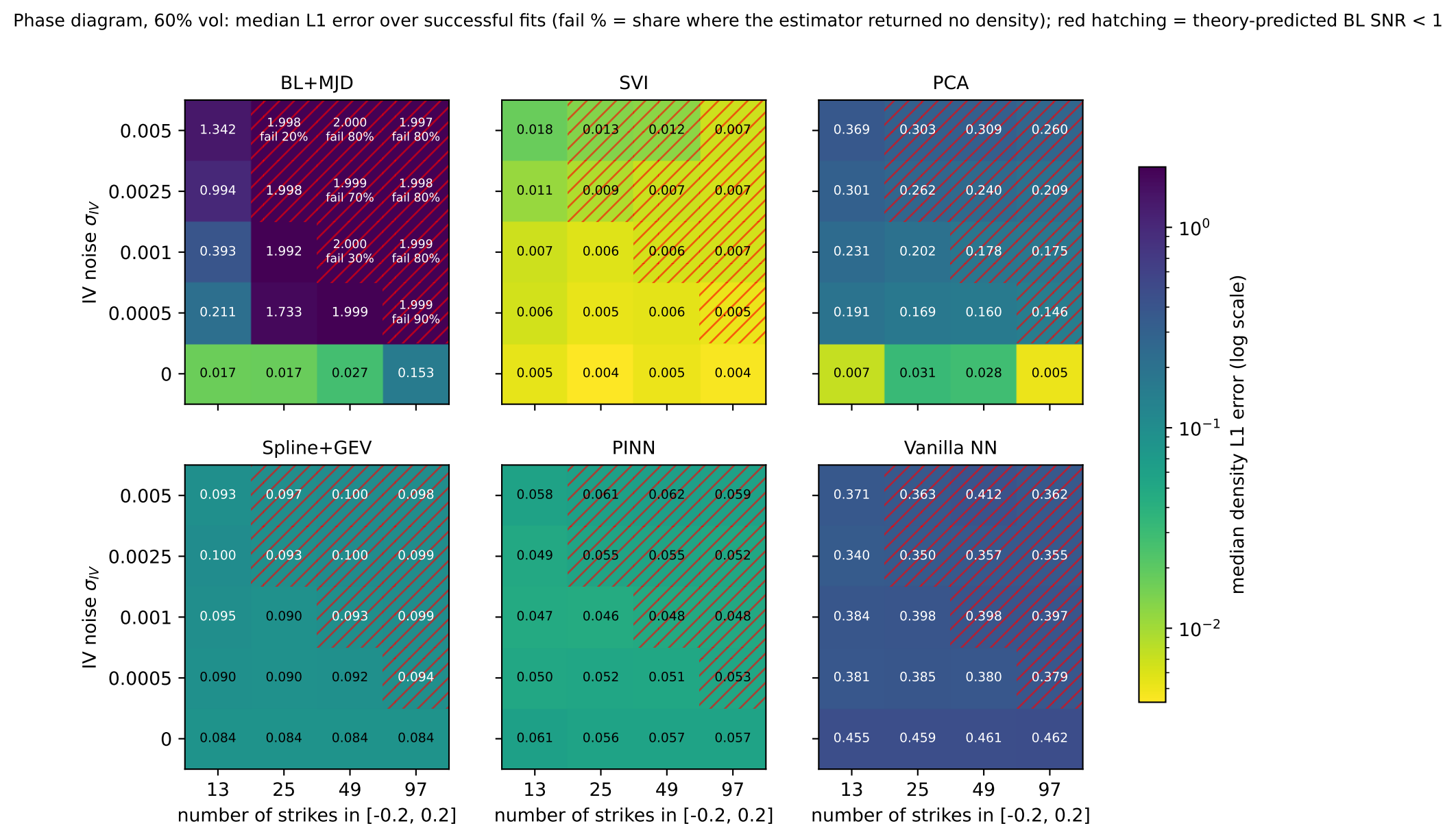
Phase diagram, 60% vol: median L1 error over successful fits (fail % = share where the estimator returned no density); red hatching = theory-predicted BL SNR < 1
BL+MJD
1.342
1.998 fail 20%
2.000 fail 80%
1.997 fail 80%
0.994
1.998
1.999 fail 70%
1.998 fail 80%
0.393
1.992
2.000 fail 30%
1.999 fail 80%
0.211
1.733
1.999
1.999 fail 90%
0.017
0.017
0.027
0.153
SVI
0.018
0.013
0.012
0.007
0.011
0.009
0.007
0.007
0.007
0.006
0.006
0.007
0.006
0.005
0.006
0.005
0.005
0.004
0.005
0.004
PCA
0.369
0.303
0.309
0.260
0.301
0.262
0.240
0.209
0.231
0.202
0.178
0.175
0.191
0.169
0.160
0.146
0.007
0.031
0.028
0.005
Spline+GEV
0.093
0.097
0.100
0.098
0.100
0.093
0.100
0.099
0.095
0.090
0.093
0.099
0.090
0.090
0.092
0.094
0.084
0.084
0.084
0.084
PINN
0.058
0.061
0.062
0.059
0.049
0.055
0.055
0.052
0.047
0.046
0.048
0.048
0.050
0.052
0.051
0.053
0.061
0.056
0.057
0.057
Vanilla NN
0.371
0.363
0.412
0.362
0.340
0.350
0.357
0.355
0.384
0.398
0.398
0.397
0.381
0.385
0.380
0.379
0.455
0.459
0.461
0.462
IV noise σ_{IV}
0.005
0.0025
0.001
0.0005
0
13
25
49
97
number of strikes in [-0.2, 0.2]
median density L1 error (log scale)
10^{0}
10^{-1}
10^{-2}